

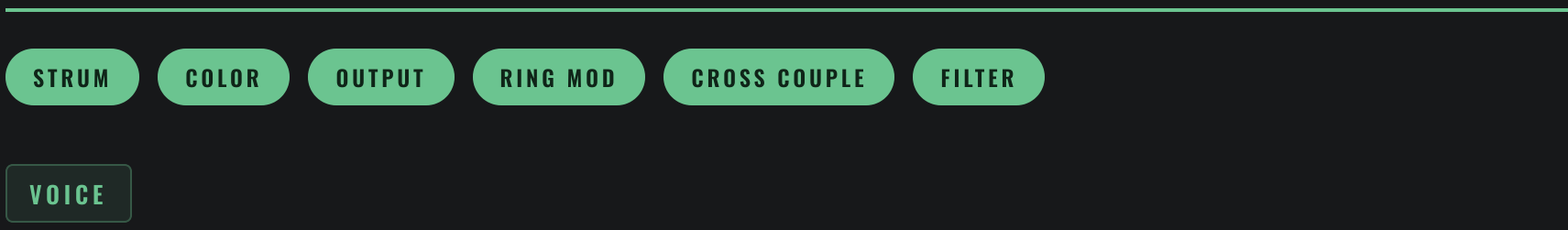


01 What It Is

Strike is a physically-modeled string synth built on the extended Karplus-Strong algorithm: a delay line seeded with a burst of excitation, feeding back through a loop filter that shapes decay and timbre. Pluck/Bow morphs the excitation continuously from a single plucked hit to a continuously bowed, sustained tone.

02 Signal Path

Excitation feeds the string loop, which recirculates through Waveshape and Filter before Ring Mod and Output shape the tone leaving it. Post Filter swaps whether Waveshape or Filter comes first. Cross-Couple (Dual topology only) links a second string loop into the first.



03 The Panel

STRUM · COLOR · OUTPUT · RING MOD · CROSS COUPLE · FILTER

VOICE



Mono on/off · default off

Off runs the original 8-voice polyphonic pool. On drives a single voice with last-note priority: holding two notes sounds only the most recent, and releasing it retriggers whichever earlier note is still held.

STRUM



Decay 0–100% · default 60%

Loop gain, controlling how long the string rings. Decay time is pitch-dependent at a fixed setting: higher notes decay faster in real time than lower ones.



Brightness 0–100% · default 100%

Tone of the plucking excitation. 0% is heavily lowpassed noise, 100% is raw white noise. Only takes effect on the next pluck.



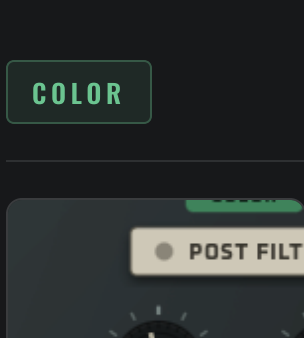
Pluck / Bow 0–100% · default 0%

Morphs the excitation from a single plucked hit (0%) toward a continuously bowed, sustained tone. Live-adjustable, unlike Brightness.



Bow Force 0–100% · default 50%

The bow's own pressure against the string. Has no effect at Pluck/Bow=0%.



Noise Color Cold / Warm / Dark · default Cold

The plucking excitation's own spectral color, from a flat, full-spectrum noise source (Cold) to a progressively darker one (Dark).

COLOR



Structure 0–100% · default 0%

Inharmonicity: stretches the string's partials sharp of exact integer multiples of the fundamental, the way a real stiff string (piano, bell-like material) does. 0% is a bit-exact no-op; higher settings sound progressively more metallic and, above about 75%, genuinely unstable in pitch.



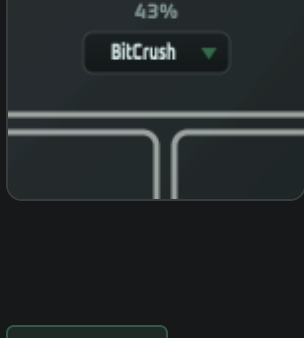
Position 0–50% · default 50%

Where the string is excited and listened to along its length. Has no neutral setting; every value changes the tone. The default (the midpoint) cancels every even harmonic for a hollow, more open character.



Post Filter Pre Filter / Post Filter · default Pre Filter

Places Waveshape's distortion before or after the Filter section's own coloring, in the output signal only.



Waveshape 0–100% · default 0%

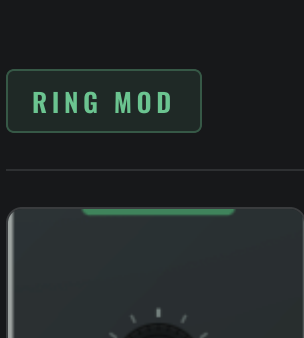
Amount of whichever Waveshaper Type is selected, applied inside the feedback loop so the distortion becomes part of what the string is resonating with. 0% is a bit-exact no-op.



Waveshaper Type Fold / BitCrush · default Fold

Fold reflects the signal back on itself past a threshold for dense, often inharmonic overtones. BitCrush reduces sample rate and bit depth for a lo-fi, digital character.

OUTPUT



Volume -60 to +6 dB · default -6 dB

Output gain, applied after all 8 voices are summed.

RING MOD



Amount 0–100% · default 0%

Depth of the in-loop ring modulator. Multiplies the loop signal by a sine oscillator at Frequency. 0% is a bit-exact no-op.



Frequency 20 Hz–5 kHz · default 200 Hz

The ring modulator's own oscillator frequency. Deliberately doesn't track the note's pitch, which is what produces ring mod's characteristic inharmonic sidebands rather than a harmonic effect.

CROSS COUPLE



On Single / Dual · default Single

Switches between a single string line (off) and two independently-excited lines cross-coupled together (on), modeling the double-decay/beating character of real coupled piano unison strings.



Amount 0–100% · default 0%

Dual topology only. How much of each line's signal feeds into the other.



Delay 0–10 ms · default 0 ms

Dual topology only. A short delay inserted into the cross-coupling path itself, turning the coupling from a flat, broadband effect into a harmonic-dependent one.



Detune 0–100% · default 0%

Dual topology only. Offsets the second line's pitch from the first by up to about 50 cents at 100%. Set once per note, not live, the same as Brightness.

FILTER



On Two-Point Average / Resonant · default Two-Point Average

Switches the loop filter from the classic Karplus-Strong averager (off) to a resonant lowpass cascaded on top of it (on), unlocking Cutoff and Resonance below.



Cutoff 20 Hz–18 kHz · default 8.0 kHz

The lowpass filter's own cutoff frequency, setting where the highs taper off of the signal.



Resonance 0–100% · default 0%

How much of the resonant peak is mixed in. Also drives the peak's Q, for a narrower, ringier resonance at higher settings.



Attack 1–1000 ms · default 1 ms

How quickly the Envelope reaches its full sweep after a note starts.



Envelope -100 to +100% · default 0%

Controls the amount of change that the filter goes through, and in which direction. 0% means nothing happens. To the left pulls the cutoff down on each note, to the right pulls the cutoff higher.



Decay 5–5000 ms · default 200 ms

How quickly the Envelope's sweep settles back after its attack.

